

Python - Unpack Tuple Items

Unpack Tuple Items

The term "unpacking" refers to the process of parsing **tuple** items in individual **variables**. In Python, the parentheses are the default delimiters for a literal representation of sequence object.

Following statements to declare a tuple are identical.

```
>>> t1 = (x,y)
>>> t1 = x,y
>>> type (t1)
<class 'tuple'>
```

Example

To store tuple items in individual variables, use multiple variables on the left of assignment operator, as shown in the following example –

```
tup1 = (10,20,30)
x, y, z = tup1
print ("x: ", x, "y: ", "z: ",z)
```

It will produce the following **output** –

```
x: 10 y: 20 z: 30
```

That's how the tuple is unpacked in individual variables.

In the above example, the number of variables on the left of assignment operator is equal to the items in the tuple. What if the number is not equal to the items?

ValueError While Unpacking a Tuple

If the number of variables is more or less than the length of tuple, Python raises a `ValueError`.

Example



```
tup1 = (10,20,30)
x, y = tup1
x, y, p, q = tup1
```

It will produce the following **output** –

```
x, y = tup1
^^^^
ValueError: too many values to unpack (expected 2)
x, y, p, q = tup1
^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 4, got 3)
```

Unpack Tuple Items Using Asterisk (*)

In such a case, the "*" symbol is used for unpacking. Prefix "*" to "y", as shown below –

Example 1



```
tup1 = (10,20,30)
x, *y = tup1
print ("x: ", "y: ", y)
```

It will produce the following **output** –

```
x: y: [20, 30]
```

The first value in tuple is assigned to "x", and rest of items to "y" which becomes a list.

Example 2

In this example, the tuple contains 6 values and variables to be unpacked are 3. We prefix "*" to the second variable.

```
tup1 = (10,20,30, 40, 50, 60)
x, *y, z = tup1
print ("x: ",x, "y: ", y, "z: ", z)
```

It will produce the following **output** –

```
x: 10 y: [20, 30, 40, 50] z: 60
```

Here, values are unpacked in "x" and "z" first, and then the rest of values are assigned to "y" as a list.

Example 3

What if we add "*" to the first variable?

```
tup1 = (10,20,30, 40, 50, 60)
*x, y, z = tup1
print ("x: ",x, "y: ", y, "z: ", z)
```

It will produce the following **output** –

```
x: [10, 20, 30, 40] y: 50 z: 60
```

Here again, the tuple is unpacked in such a way that individual variables take up the value first, leaving the remaining values to the list "x".

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